Programming and Data Structures Laboratory (CS19003)
Assignment 3: Section 6
Department of CSE, IIT Kharagpur

28th January, 2026

General instructions to be followed strictly:

- Name your file as <roll_no>_<assignment_no>_<question_no>. For example, if your roll number is `14CS10001` and you are submitting assignment 3 question number (a), then name your file as `14CS10001_3_a.c`
- Write your name, roll number, and assignment number at the beginning of your program.
- Use proper indentation in your code and comment. Marks may be deducted without proper indentation and comments.
- Submitting the **correct file** is your responsibility, any request of re-submission after the lab will **never** be entertained.
- **Zero tolerance** policy will be stricly followed for **plagiarism**. The marks will be put straight zero if such cases are identified.

Question-1 (10 points)

Write a program that takes an integer between 1 and 9 as input, and prints out on the terminal a pattern similar to the following (shown for input 5).
Example:

```
Enter an integer between 1 and 9: 5
1                 1
2 2             2 2
3 3 3         3 3 3
4 4 4 4     4 4 4 4
5 5 5 5 5 5 5 5 5 5
```

Question-2 (20 points)

Accept from the user a non-negative integer r. Consider the equation $x + y + z = r$ where x, y, z are the unknown variables. Write a C program to find the non-negative integer solutions for x, y, z satisfying this equation. Display all of them. For example, if $r = 2$, the possible solutions are $(0, 0, 2), (0, 1, 1), (0, 2, 0), (1, 0, 1), (1, 1, 0), (2, 0, 0)$. An example run of your program can be:

```
Enter non-negative integer r: 3
x, y, z values are: (0, 0, 3) (0, 1, 2) (0, 2, 1) (0, 3, 0) (1, 0, 2) (1, 1, 1) (1, 2, 0) (2, 0, 1)

(2, 1, 0) (3, 0, 0)
```

Question-3 (20 points)

The Fibonacci Series is the series of numbers: $0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \cdots$. The starting numbers are 0 and 1, and the next numbers in the series are found by adding up the two numbers before it. Write a C program to print the numbers that **do not** appear in the Fibonacci series. The user will input the number of terms in the Fibonacci sequence to be considered. For example, if the user enters $n = 7$, the values to be printed are: 4 6 7.